

Low-Complexity Near-Field Localization with XL-MIMO Sectored Uniform Circular Arrays

Shicong LIU and Xianghao YU

Dept. Electrical Engineering, City University of Hong Kong, Hong Kong SAR

Slides & Codes: <https://github.com/scliubit/XL-MIMO-sUCA-Loc>

December 11, 2024



香港城市大學
City University of Hong Kong



NSFC
National Natural Science
Foundation of China



大學教育資助委員會
University Grants Committee



IEEE
GLOBECOM®

IEEE
ComSoc®



Outlines

- ➊ Introduction
- ➋ System Model
- ➌ Method and Performance
 - Backprojection
 - Performance
- ➍ Simulations
- ➎ References

Outline

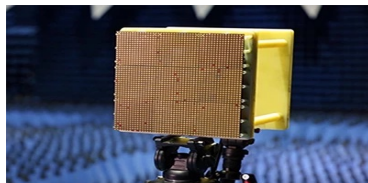
- 1 **Introduction**
- 2 System Model
- 3 Method and Performance
 - Backprojection
 - Performance
- 4 Simulations
- 5 References

Introduction

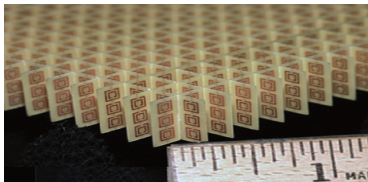
- XL-MIMO antenna arrays have been widely investigated in recent years



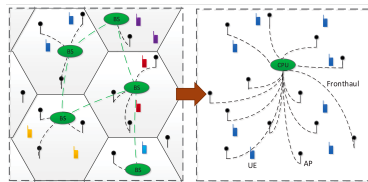
(a) Massive MIMO BS



(b) Reconfigurable Intelligent Surface



(c) Holographic MIMO



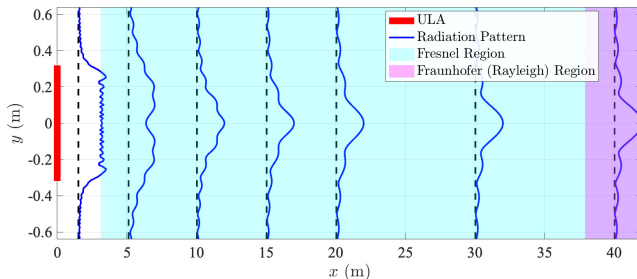
(d) Cell-free mMIMO

Figure 1: Widely investigated XL-MIMO related topics.

Introduction

- Proliferation of antennas (in XL-MIMO) results in increased **aperture**, together with smaller **wavelength** extends the **near-field region** [1]

$$\sqrt[3]{\frac{2D^4}{\lambda}} \leq d_{\text{NF}} \leq \frac{2D^2}{\lambda}, \quad D \uparrow \quad \lambda \downarrow \Rightarrow d_{\text{NF}} \uparrow$$



- Distance-dependent beam pattern poses challenges for wireless systems in the near field

[1] J. Goodman, *Introduction to Fourier Optics* (McGraw-Hill physical and quantum electronics series). W. H. Freeman, 2005, ISBN: 9780974707723.

- Distance r and Angle θ determine near-field channel together

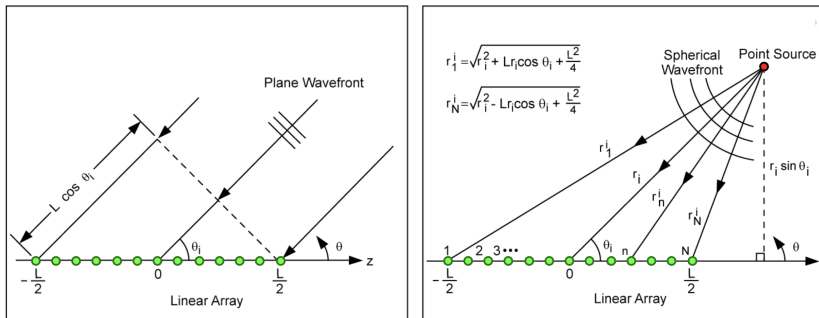
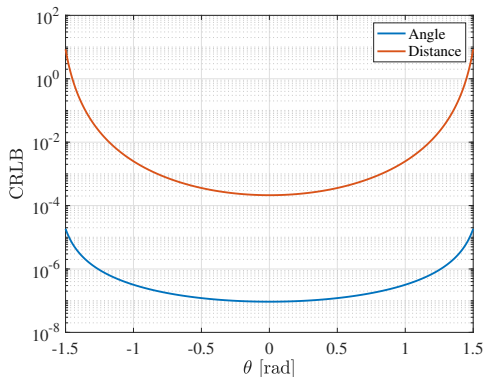


Figure 2: Wavefront illustration [2].

[2] A. Fenn, "Evaluation of adaptive phased array antenna, far-field nulling performance in the near-field region," *IEEE Trans. Antennas Propag.*, vol. 38, no. 2, pp. 173–185, 1990.

Introduction

- Linear (Planar) Arrays are not good at sensing
 - ▶ Resolution drains at **edges**.
 - ▶ **Range** sensing performance worse than angle's [3].



[3] H. Wang, Z. Xiao, and Y. Zeng, "Cramér-rao bounds for near-field sensing with extremely large-scale MIMO," *IEEE Transactions on Signal Processing*, vol. 72, pp. 701–717, 2024.

Previous Works

Emerging works for utilizing circular arrays:

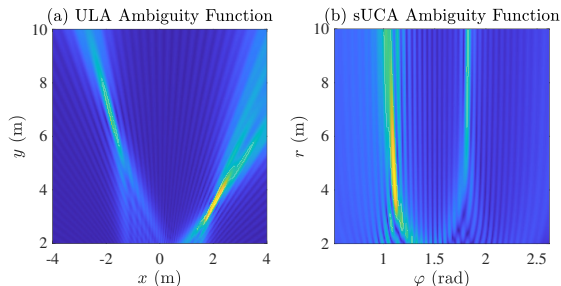
- Uniform Circular Array (UCA) for azimuthally consistent resolution [4].
- Utilizing UCA for reduced codebook size [5].
- Rotational symmetry of UCA provides uniform and enlarged near-field regions at all angles [6].
- Near-field beamspace beamforming with UCA for better beam pattern [7].

-
- [4] T. Liang, M. Zhu, and F. Pan, "A DOA estimation method for uniform circular array based on virtual interpolation and subarray rotation," *IEEE Access*, vol. 9, pp. 116 760–116 767, 2021.
- [5] Y. Xie, B. Ning, *et al.*, "Near-field beam training in thz communications: The merits of uniform circular array," *IEEE Wireless Commun. Letters.*, vol. 12, no. 4, pp. 575–579, 2023.
- [6] Z. Wu, M. Cui, and L. Dai, "Enabling more users to benefit from near-field communications: From linear to circular array," *IEEE Trans. Wireless Commun.*, vol. 23, no. 4, pp. 3735–3748, 2024.
- [7] F. Zhang and W. Fan, "Near-field ultra-wideband mmwave channel characterization using successive cancellation beamspace uca algorithm," *IEEE Trans. Veh. Technol.*, vol. 68, no. 8, pp. 7248–7259, 2019.

Previous Works

Our motivations:

- Better beam pattern representation for localization



- Consistent resolution with improvements in distance
- Fast positioning with FFT

Outline

- 1 Introduction
- 2 System Model**
- 3 Method and Performance
 - Backprojection
 - Performance
- 4 Simulations
- 5 References

System Model

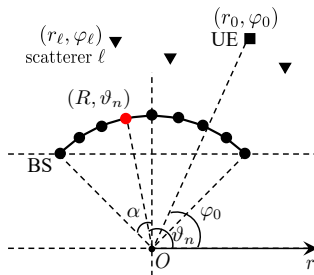


Figure 3: Sectored uniform circular array (sUCA) with N antennas.

- Near-field received signal at the k -th subcarrier f_k

$$\mathbf{h}_k[n] = \frac{1}{\sqrt{NL}} \sum_{\ell=0}^L \frac{\alpha_{\ell} e^{j \frac{2\pi}{c} f_k (d_{\ell} + d_{\ell,n})}}{d_{\ell} d_{\ell,n}} = \sum_{\ell=0}^L \frac{\tilde{\alpha}_{k,\ell} e^{j \frac{2\pi}{c} f_k d_{\ell,n}}}{d_{\ell,n}}, \quad (1)$$

where

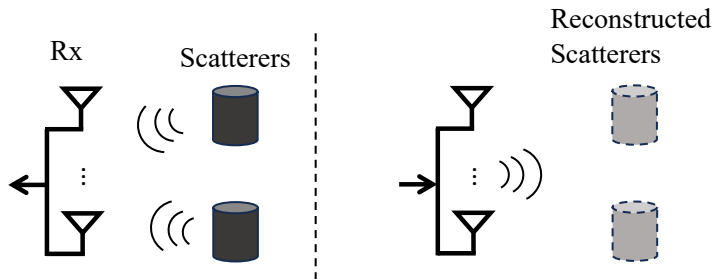
$$d_{\ell,n} = \sqrt{r_{\ell}^2 + R^2 - 2Rr_{\ell} \cos(\vartheta_n - \varphi_{\ell})} \approx r_{\ell} + \frac{R^2}{2r_{\ell}} - R \cos(\vartheta_n - \varphi_{\ell}) \quad (2)$$

Outline

- 1 Introduction
- 2 System Model
- 3 Method and Performance**
 - **Backprojection**
 - **Performance**
- 4 Simulations
- 5 References

Backprojection

- Backprojection algorithm reconstruction the target by emulating the back propagation of the EM wave

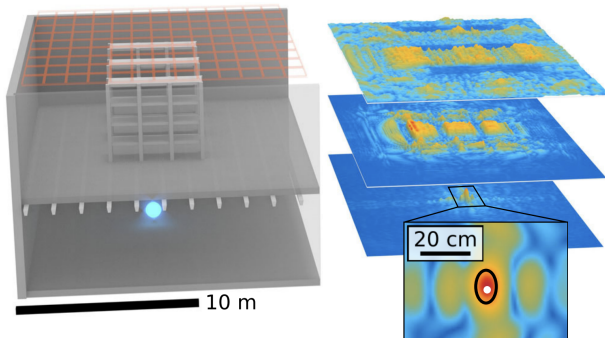


(a) Recording Wavefront

(b) **Emulate** Wavefront Back-projection

Backprojection

- It has been widely adopted in sensing/imaging applications [8]



[8] P. M. Holl and F. Reinhard, "Holography of wi-fi radiation," *Phys. Rev. Lett.*, vol. 118, p. 183 901, 18 May 2017.

Backprojection Reconstruction

- Emulated backprojection signal, a.k.a., the *ambiguity function*

$$\begin{aligned}
 F_k(r, \varphi) &= \sum_{n=1}^N \underbrace{\mathbf{y}_k[n] s_k^*}_{\text{Wavefront}} \underbrace{e^{-j \frac{2\pi}{c} f_k d_n}}_{\text{Propag.}} = \sum_{\ell=0}^L \sum_{n=1}^N \frac{\tilde{\alpha}_{k,\ell} e^{j \frac{2\pi}{c} f_k (d_{\ell,n} - d_n)}}{d_{\ell,n}} + \tilde{\mathbf{n}}_k \\
 &\triangleq \sum_{\ell=0}^L F_k^{(\ell)}(r, \varphi) + \tilde{\mathbf{n}}_k
 \end{aligned} \tag{3}$$

at arbitrary coordinate (r, φ) and the k -th subcarrier $f_k = f_c + (k/K)f_{\text{SCS}}$.

- Usually, subcarrier spacing $f_{\text{SCS}} = 15 \text{ kHz}$
- We proved two basic propositions regarding $F_k^{(\ell)}(r, \varphi)$ in (3)

Propositions

Proposition 1

There exists only one unique peak for each $F_k^{(\ell)}(r, \varphi)$ at corresponding (r_ℓ, φ_ℓ)

$$\left| F_k^{(\ell)}(r, \varphi) \right|^2 \stackrel{(a)}{\leq} \sum_{n=1}^N \frac{\left| e^{j\frac{4\pi}{c} f_k R \cos(\vartheta_n - \varphi_\ell)} \right|^2}{\hat{d}_{\ell,n}} \sum_{n=1}^N \frac{\left| e^{-j\frac{4\pi}{c} f_k R \cos(\vartheta_n - \varphi)} \right|^2}{\hat{d}_{\ell,n}} \quad (4)$$

Proposition 2

For points $(r, \varphi) \neq (r_\ell, \varphi_\ell)$, $F_k^{(\ell)}(r, \varphi)$ vanishes asymptotically as $f_k \rightarrow \infty$ and $N \rightarrow \infty$

Demonstration

■ Demonstration of the two propositions.

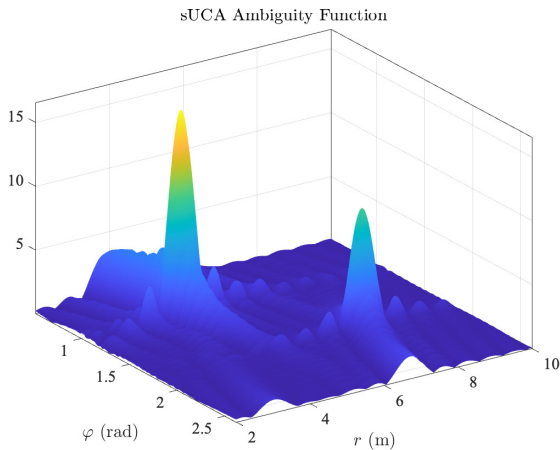


Figure 4: Two targets located at $(4, \frac{\pi}{3})$ and $(7, \frac{7\pi}{12})$, respectively.

FFT Implementation

- Notice that for given distance r we can rewrite (3) as

$$F_k(\varphi \mid r, r_\ell, \varphi_\ell) = \sum_{n=1}^N \mathbf{y}_k[n] e^{-j \frac{2\pi}{c} f_k \sqrt{r^2 + R^2 - 2Rr \cos(\vartheta_n - \varphi)}}. \quad (5)$$

- Define

$$\bar{\mathbf{e}}_{r(j)} = \left[e^{j \frac{2\pi f_k}{c} \sqrt{\tilde{r} - 2Rr \cos(\varphi^{(1)})}}, \dots, e^{j \frac{2\pi f_k}{c} \sqrt{\tilde{r} - 2Rr \cos(\varphi^{(G_a)})}} \right], \quad (6)$$

the near-field reconstruction can be implemented by FFT as

$$\Theta_k[:, j] = \mathcal{F}_{G'_a}^{-1} [\mathcal{F}_{G'_a} [\bar{\mathbf{y}}_k] \odot \mathcal{F}_{G'_a} [\bar{\mathbf{e}}_{r(j)}]]_{N:G'_a-N+1}, \quad (7)$$

Performance

In short, with reasonable approximations,

- Angular resolution (mainlobe width)

$$\Delta\varphi = \frac{\lambda}{2R \sin \alpha} = \frac{\lambda}{2R} \cdot \frac{1}{\sin \alpha}, \quad (8)$$

whereas ULA:

$$\Delta\varphi = \frac{2\lambda}{\pi R \cos \theta} = \frac{\lambda}{2R} \cdot \frac{4}{\pi \cos \theta} \quad (9)$$

- Distance resolution (mainlobe width)

$$\Delta r = \frac{2c}{f_{\text{BW}}}, \quad (10)$$

whereas ULA beampattern does not show zero-crossing points in the near-field.

Algorithm

Require: $\{\mathbf{y}_k\}_{k=1}^K$, grids G_a and G_d , α , distance grid range $[r_{\min}, r_{\max}]$.

Ensure: The estimated coordinates of the UE and scatterers.

- 1: **for** $k = 1, \dots, K$ **do**
- 2: Initialize empty matrix $\Theta_k \in \mathbb{R}^{G_a \times G_d}$.
- 3: **for** $j = 1, \dots, G_d$ **do**
- 4: Fill the j -th column of Θ_k according to (7).
- 5: **end for**
- 6: Normalize Θ_k with $\Theta_k := \Theta_k / \max(|\Theta_k|)$.
- 7: **end for**
- 8: Find top $L + 1$ indices of peaks $\{i_\ell\}_{\ell=0}^L$ from

$$\Theta_a = \sum_{k=1}^K \sum_{j=1}^{G_d} |\Theta_k[:, j]|.$$

- 9: Find peak on distance grids from

$$j_\ell = \arg \max_j \left| \sum_{k=1}^K \Theta_k[i_\ell, j] \right|.$$

- 10: Angle estimations $\hat{\varphi}_\ell = \pi/2 - \alpha + 2\alpha i_\ell / G_a$.
- 11: Distance estimations $\hat{r}_\ell = r_{\min} + (r_{\max} - r_{\min}) j_\ell / G_d$.

Outline

- 1 Introduction
- 2 System Model
- 3 Method and Performance
 - Backprojection
 - Performance
- 4 Simulations**
- 5 References

Simulation Setup

■ Evaluation criteria

$$r_{\text{err}} = \frac{1}{L+1} \sum_{\ell=0}^L \sqrt{\hat{r}_{\ell}^2 + r_{\ell}^2 - 2\hat{r}_{\ell}r_{\ell} \cos(\hat{\varphi}_{\ell} - \varphi_{\ell})},$$

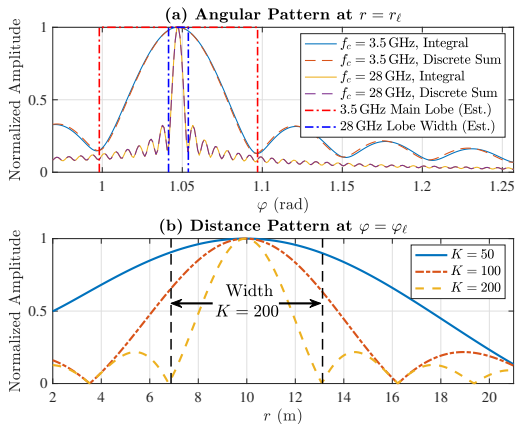
■ Sector size $2\alpha = \frac{2\pi}{3}$, near field range [2 m, 21 m]

■ $f_c = \{3.5, 10, 28\}$ GHz

■ Number of subcarriers $K = \{50, 100, 200\}$

Numerical Results

Reconstruction pattern

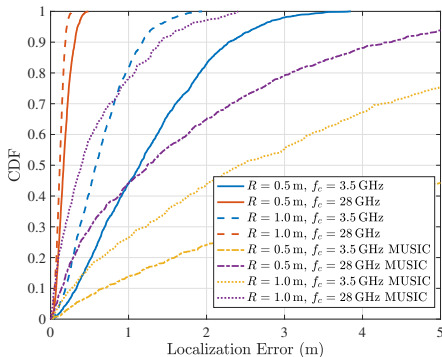


Simulation and Theoretical Results are Consistent

Numerical Results

MUSIC performance is affected due to

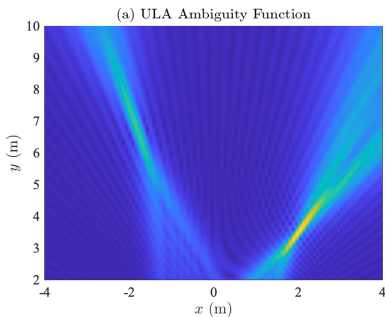
- Steering matrix of sUCA no longer satisfies Vandermonde form
- Two parameters mutually coupled in the steering vector
- Original MUSIC is not an asymptotically consistent estimator for large N



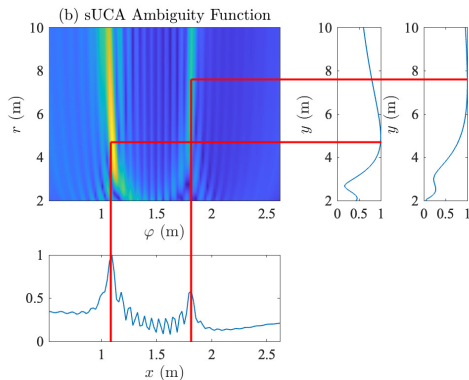
Outperforms MUSIC in sUCA

Example

- Localization can now be achieved by successive 1-D grid search with lower complexity



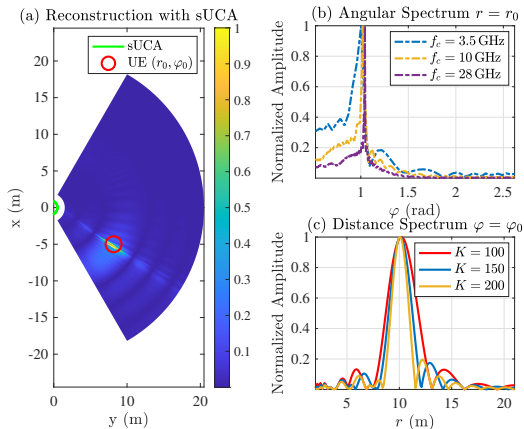
(a) ULA Beampattern



(b) sUCA Beampattern

Numerical Results

■ Demonstration on near-field reconstruction



Numerical Results

■ Complexity analysis

	Number of Multiplications	Avg. Runtime (s)	
		3.5 GHz	28 GHz
MUSIC	$KN^2 + N^3 + G_a G_d ((N - L + 1)N^2 + N)$	0.5071	58.8952
Proposed	$KG_d(G_a + 1)N$	0.5963	0.7995
Proposed (FFT)	$KG_d \lceil \log_2 G_a \rceil 2^{\lceil \log_2 G_a \rceil}$	0.0787	0.5879

Proposed Method Shows Lowest Complexity

Summary

■ Resorting to BP algorithm for near-field localization

- ▶ Arrays with **Linear** geometry are not suitable for sensing, for its performance drain at edges and worse distance resolution.
- ▶ **Circular** array, as an alternative for ULA, performs better with constant resolution and lower algorithm complexity.
- ▶ Simulations validate and demonstrate our perspectives in near-field sensing.

Outline

- 1 Introduction
- 2 System Model
- 3 Method and Performance
 - Backprojection
 - Performance
- 4 Simulations
- 5 References**

- [1] J. Goodman, *Introduction to Fourier Optics* (McGraw-Hill physical and quantum electronics series). W. H. Freeman, 2005, ISBN: 9780974707723.
- [2] A. Fenn, "Evaluation of adaptive phased array antenna, far-field nulling performance in the near-field region," *IEEE Trans. Antennas Propag.*, vol. 38, no. 2, pp. 173–185, 1990.
- [3] H. Wang, Z. Xiao, and Y. Zeng, "Cramér-rao bounds for near-field sensing with extremely large-scale MIMO," *IEEE Transactions on Signal Processing*, vol. 72, pp. 701–717, 2024.
- [4] T. Liang, M. Zhu, and F. Pan, "A DOA estimation method for uniform circular array based on virtual interpolation and subarray rotation," *IEEE Access*, vol. 9, pp. 116 760–116 767, 2021.
- [5] Y. Xie, B. Ning, L. Li, and Z. Chen, "Near-field beam training in thz communications: The merits of uniform circular array," *IEEE Wireless Commun. Letters.*, vol. 12, no. 4, pp. 575–579, 2023.
- [6] Z. Wu, M. Cui, and L. Dai, "Enabling more users to benefit from near-field communications: From linear to circular array," *IEEE Trans. Wireless Commun.*, vol. 23, no. 4, pp. 3735–3748, 2024.

- [7] F. Zhang and W. Fan, "Near-field ultra-wideband mmwave channel characterization using successive cancellation beamspace uca algorithm," *IEEE Trans. Veh. Technol.*, vol. 68, no. 8, pp. 7248–7259, 2019.
- [8] P. M. Holl and F. Reinhard, "Holography of wi-fi radiation," *Phys. Rev. Lett.*, vol. 118, p. 183901, 18 May 2017.
- [9] M. Desai and W. Jenkins, "Convolution backprojection image reconstruction for spotlight mode synthetic aperture radar," *IEEE Trans. Image Process.*, vol. 1, no. 4, pp. 505–517, 1992.
- [10] S. Li, S. Wang, M. G. Amin, and G. Zhao, "Efficient near-field imaging using cylindrical mimo arrays," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 57, no. 6, pp. 3648–3660, 2021.
- [11] D. Gabor, "Microscopy by Reconstructed Wave-Fronts," *Proceedings of the Royal Society of London Series A*, vol. 197, no. 1051, pp. 454–487, 1949.
- [12] S. Liu, X. Yu, Z. Gao, J. Xu, D. W. K. Ng, and S. Cui, "Sensing-enhanced channel estimation for near-field XL-MIMO systems," *IEEE J. Sel. Areas Commun.*, 2024, to appear.



Thanks for your attention! Q & A

Contact: sc.liu@my.cityu.edu.hk

Backprojection

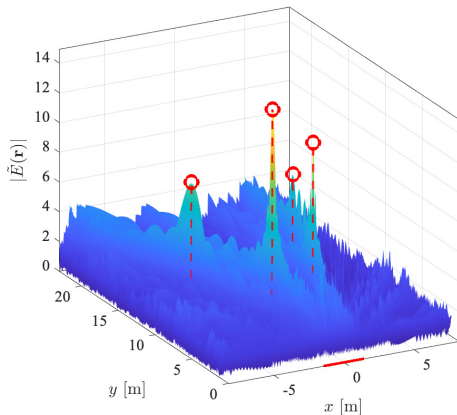
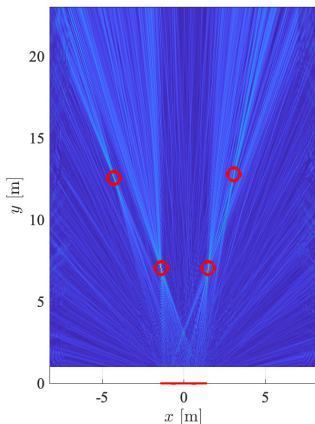
- Backprojection (BP) algorithm has been widely adopted in image reconstruction tasks [9], [10].
- It propagates the received wave front in space according to the EM wave diffraction or angular-spectrum relation [11].
- Example: Reconstruction from z_0 plane:

$$I(x, y, z) = \mathcal{F}^{-1} \left[\underbrace{e^{j \frac{2\pi(z-z_0)}{\lambda}} \sqrt{1 - \lambda^2 f_x^2 - \lambda^2 f_y^2}}_{\mathcal{F}[\text{EM Prop Kernel}]} \times \underbrace{\mathcal{F}[I_0(x, y, z_0)]}_{\text{Wavefront}} \right] \quad (11)$$

-
- [9] M. Desai and W. Jenkins, "Convolution backprojection image reconstruction for spotlight mode synthetic aperture radar," *IEEE Trans. Image Process.*, vol. 1, no. 4, pp. 505–517, 1992.
- [10] S. Li, S. Wang, *et al.*, "Efficient near-field imaging using cylindrical mimo arrays," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 57, no. 6, pp. 3648–3660, 2021.
- [11] D. Gabor, "Microscopy by Reconstructed Wave-Fronts," *Proceedings of the Royal Society of London Series A*, vol. 197, no. 1051, pp. 454–487, 1949.

Backprojection

■ Reconstruction example [12]



[12] S. Liu, X. Yu, *et al.*, "Sensing-enhanced channel estimation for near-field XL-MIMO systems," *IEEE J. Sel. Areas Commun.*, 2024, to appear.

Reference Wave

- Obtain precise coherent reference wave (carrier) by phase lock loop (PLL)

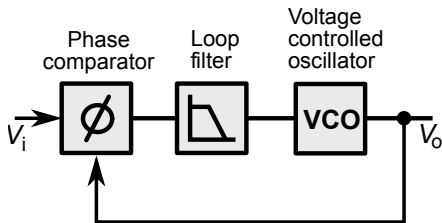


Figure 7: Simple analog phase locked loop